



Final Exam Calculus 2018 2019 Sem 2

CALCULUS - foundation (Universiti Teknologi Malaysia)



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QUESTION 1 (15 MARKS)

a) A function $f(x)$ is defined by

$$f(x) = \begin{cases} ax^2 + \frac{1}{2}, & x < -1, \\ 1 - bx, & -1 \leq x < 2, \\ \sin\left(\frac{\pi}{2}x\right), & x \geq 2, \end{cases}$$

where a and b are constants. Find the values of a and b if $f(x)$ is continuous at $x = -1$ and $x = 2$.

(5 marks)

b) Find the limit of each of the following

i. $\lim_{x \rightarrow 4} \frac{x^2 - 3x - 4}{x^2 - 16}.$

(5 marks)

ii. $\lim_{x \rightarrow \infty} \frac{4x - 1}{\sqrt{9x^2 + 1} + 2x}.$

(5 marks)

QUESTION 2 (15 MARKS)

a) If $y = \sec^2(1 - x^2)$, find $\frac{dy}{dx}$.

(3 marks)

b) The parametric equations of a curve is given by

$$x = t^2 + t \text{ and } y = t^3 - t.$$

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t .

(5 marks)

c) If $y = f(x)$ is given implicitly as

$$x^2 - xy + y^2 = 9,$$

show that

$$(2y - x)\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 - 2\frac{dy}{dx} + 2 = 0.$$

Hence, find the value of $\frac{d^2y}{dx^2}$ if $x = 0$ and $y > 0$.

(7 marks)

QUESTION 3 (15 MARKS)

a) The derivative of a function $y = f(x)$ is given as

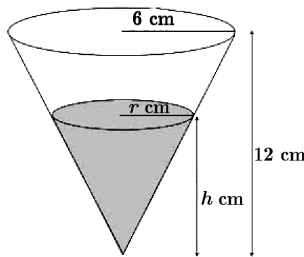
$$\frac{dy}{dx} = \frac{x^2 - 4x - 5}{x - 2}.$$

i. Find all the critical values of the function.

(4 marks)

ii. Use First Derivative Test to determine which critical values will give maximum or minimum points.

(4 marks)



b) An empty cone of radius 6 cm and height 12 cm is being filled with water. At certain instant, the radius of water is r cm and height h cm.

i. Show that the volume of water is

$$V = \frac{\pi}{12}h^3.$$

(3 marks)

ii. If the volume flow rate of water is $20 \text{ cm}^3 \text{ s}^{-1}$, find the rate of water level is rising when $h = 2$ cm.

(4 marks)

QUESTION 4 (18 MARKS)

a) Evaluate each of the following integrals:

i. $\int (\sin 5x \cos 3x + \cos^2 x) dx.$

(5 marks)

ii. $\int e^{2x} \cos 3x dx.$

(5 marks)

b) Using the substitution $u = e^x$, show that

$$\int \frac{e^{2x}}{e^{2x} - e^x - 2} dx = \int \frac{u}{u^2 - u - 2} du.$$

Hence evaluate

$$\int \frac{e^{2x}}{e^{2x} - e^x - 2} dx.$$

(8 marks)

QUESTION 5 (10 MARKS)

a) The area of the region bounded by the curve $y = ax^2$ where $a > 0$, lines $x = 2$ and $y = -x$ is given as 10 square unit. Find the value of a .

(5 marks)

b) The region bounded by the curve $y = \frac{1}{x^2}$, lines $y = 8x$ and $y = 1$ is revolved about the y -axis. Find the volume of the resulting solid.

(5 marks)

QUESTION 6 (10 MARKS)

Solve each of the following differential equations which satisfy the given initial conditions.

a) $2y(x+2)\frac{dy}{dx} = 4 + y^2$, given $y(2) = 2$.

(5 marks)

b) $x\frac{dy}{dx} - y = x^2e^{-2x}$ with $y(1) = 2$.

(5 marks)

QUESTION 7 (17 MARKS)

a) Given $f(x) = x^3 - 5x - 2$.

- i. Using Intermediate Value Theorem, verify that the equation $f(x) = 0$ has a root between 2 and 3. Then use the iterative formula

$$x_{i+1} = \frac{2x_i^3 + 2}{3x_i^2 - 5}, \quad i = 0, 1, 2, \dots,$$

starting with $x_0 = 2$, to find this root to three decimal places. Use stopping criteria $|x_{i+1} - x_i| < 0.0005$.

(6 marks)

- ii. Hence, find to two decimal places, a root of

$$2^{3x} - 5(2^x) - 2 = 0.$$

(4 marks)

- b) If x is in radians, use Simpson's rule with eight strips to obtain an estimate of

$$\int_2^3 \cos(x - 2) \ln x \, dx.$$

Give your answer correct to three decimal places.

(7 marks)

LIST OF FORMULA

Trigonometric	Differentiations	Integrations
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$	$\frac{d}{dx}[k] = 0, k \text{ constant}$ $\frac{d}{dx}[x^n] = nx^{n-1}$ $\frac{d}{dx}[e^x] = e^x$ $\frac{d}{dx}[\ln x] = \frac{1}{x}$ $\frac{d}{dx}[\cos x] = -\sin x$ $\frac{d}{dx}[\sin x] = \cos x$ $\frac{d}{dx}[\tan x] = \sec^2 x$ $\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$ $\frac{d}{dx}[\sec x] = \sec x \tan x$ $\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int k dx = kx + C, k \text{ constant}$ $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$ $\int e^x dx = e^x + C$ $\int \frac{dx}{x} = \ln x + C$ $\int \sin x dx = -\cos x + C$ $\int \cos x dx = \sin x + C$ $\int \sec^2 x dx = \tan x + C$ $\int \operatorname{cosec}^2 x dx = -\cot x + C$ $\int \sec x \tan x dx = \sec x + C$ $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$
Logarithm		
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$		

Volume of Circular Cone
$V = \frac{1}{3}\pi r^2 h$
Fixed Point Formula
$x_{i+1} = g(x_i)$
Newton-Raphson Formula
$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$
Trapezoidal rule
$\int_{x_0}^{x_N} f(x) dx = \frac{h}{2} [f_0 + f_N + 2(f_1 + f_2 + \cdots + f_{N-1})], h = \frac{x_N - x_0}{N}$
Simpson's rule
$\int_{x_0}^{x_N} f(x) dx = \frac{h}{3} [f_0 + f_N + 4(f_1 + f_3 + \cdots + f_{N-1}) + 2(f_2 + f_4 + \cdots + f_{N-2})], h = \frac{x_N - x_0}{N}$